

[0] $a^n + b^n = c^n$ where $a < b < c$ and a, b, c , and n are positive integers with $n \geq 2$
 since $c > a$ and $c > b$, c can be written as $x+a$ or $y+b$ where x and y are positive integers

[1] thus $a^n + b^n = (x+a)^n = (y+b)^n$

now expand $(x+a)^n$ and $(y+b)^n$:

$(x+a)^n = x^n + a^n + \text{remaining terms (call their sum "f")}$

$(y+b)^n = y^n + b^n + \text{remaining terms (call their sum "g")}$

thus $a^n + b^n = x^n + a^n + f = y^n + b^n + g$

subtract from both sides

$b^n = x^n + f$

$a^n = y^n + g$

f and g are both positive integers, thus $b^n > x^n$ and $a^n > y^n$

take n th root of both sides

$b > x$ and $a > y$

write b as $x+s$ and a as $y+r$ where s and r are positive integers

now back to [1]

$a^n + b^n = (x+a)^n = (y+b)^n$

$a^n + b^n = (x+(y+r))^n = (y+(x+s))^n$

$a^n + b^n = (x+y+r)^n = (x+y+s)^n$

thus $r = s$; rename this positive integer Z .

[2] then $a^n + b^n = (x+y+Z)^n$

by definition, $a \neq b$ and since $c = x+a = y+b$, $x \neq y$

[3] $y+Z = a$ and $y < a$, thus **a must be at least 2** (if $y = Z = 1$)

$b > a$ so **b must be at least 3**

[4] thus $a + b \geq 5$

now back to [2]:

$a^n + b^n = (x+y+Z)^n$

recall that $b = x+Z$ and $a = y+Z$. thus $x = b-Z$ and $y = a-Z$. Substitute these values:

$a^n + b^n = ((b-Z)+(a-Z)+Z)^n$

$a^n + b^n = (a + b - 2Z + Z)^n = (a + b - Z)^n$

Z must be at least 1, so $(a + b - Z)^n \geq (a + b - 1)^n$

$a^n + b^n \geq (a + b - 1)^n$

now let $k = a+b$. Substitute

$(k-b)^n + (k-a)^n \geq (k - 1)^n$

$(k-a)^n$ is the bigger of the two terms on the left since $a < b$.

thus $2^*(k-a)^n > (k-b)^n + (k-a)^n \geq (k - 1)^n$

recall from [3] that $a \geq 2$. then

$$2^k(k-2)^n \geq 2^k(k-a)^n > (k-b)^n + (k-a)^n \geq (k-1)^n$$

or simply

$$2^k(k-2)^n \geq (k-1)^n$$

write the factor 2 (on the left side) as $(2^{1/n})^n$

$$((2^{1/n})^n)^k (k-2)^n \geq (k-1)^n$$

take nth root of both sides

$$(2^{1/n})^k (k-2) \geq (k-1)$$

divide both sides by $(k-2)$

$$(2^{1/n})^k \geq (k-1) / (k-2)$$

recall that $k = a+b$ and, from [4], $a + b \geq 5$. the function $f(k) = (k-1) / (k-2)$ is decreasing on the interval $(5, \infty)$. thus its maximum in this interval is at $f(5) = (5-1) / (5-2) = 4/3 = 1.33333...$ Thus we can write

$$(2^{1/n})^k \geq (k-1) / (k-2) \geq 4/3$$

or simply

$$[5] \quad (2^{1/n})^k \geq 4/3$$

for $n = 2$, $(2^{1/n})^k = 1.4142... \geq 4/3$

but for $n = 3$, $(2^{1/n})^k = 1.2599...$ which is not greater than $4/3$, and the same is true for all $n > 3$ thus the original equation [0] holds only for $n = 2$

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